CASE STUDY

The SharePoint Server Exploitation Waves of 2026 (CVE-2026-45659 and Successors): A Case Study

A TeamStarWolf case study. Published 11 November 2026; every source checked as of that date.

Executive summary

The people who carried this year were the administrators of on-premises SharePoint Server farms. Between 18 March and 25 September 2026, CISA added eight 2026 SharePoint Server CVEs to its Known Exploited Vulnerabilities (KEV) catalog [3], and on 15 July SharePoint Server 2016 and 2019 reached the end of extended support [17] (**Confirmed**).

No government record opened names an actor [1] [15] [16]; CISA ties the wave to IIS machine-key theft and to deserialization used for persistence and malware [1] (**Confirmed**). The KEV entry for CVE-2026-45659 now reads Known ransomware use [3] (**Confirmed**), while Microsoft's own record for that CVE still says it is not exploited [9] (**Confirmed**).

Key facts at a glance

Field	Value
Product	SharePoint Server Subscription Edition, 2019 and 2016 [1]
CVEs in KEV (2026)	CVE-2026-20963, -32201, -45659, -56164, -58644, -50522, -55040, -65660 [3]
KEV dates	18 Mar to 25 Sep 2026 [3] [4]
Actor	None named by any government source [1] [14] [15] [16]; Symantec's Warlock reporting names no 2026 CVE [25] (**Reported**)
End of support	2016 and 2019, 15 Jul 2026 [17]

Key findings

- **KEV, not the vendor flag, carried the signal.** Four of the eight KEV-listed CVEs still show Microsoft's exploited flag as No [3] [7] [8] [9] (**Confirmed**), so KEV is the trigger to use (**Inferred**).
- **Two fixes shipped before their CVEs.** The CVE-2026-45659 fix was in the May updates and the CVE-2026-58644 fix in the June updates, before either CVE was listed [9] [10] (**Confirmed**). Programs that tracked the CVE list rather than the build had no advisory signal until the CVE appeared (**Inferred**).
- **Patching does not end the incident.** CISA says to remove machine-key harvesters, then rotate the keys [1] (**Confirmed**); stolen keys reportedly let attackers keep access after patching [22] (**Reported**).
- **Severity drifted upward.** CVE-2026-20963 became an unauthenticated 9.8 two months after its fix [8]; CVE-2026-65660 was re-titled from spoofing to code execution [11] (**Confirmed**).
- **The ransomware story is real but not yet tied to a CVE.** Symantec describes Warlock intrusions in the two months before 1 October 2026 that began with on-premises SharePoint exploitation, one of them dated 22 to 31 July, but ties no specific 2026 CVE to any intrusion [25] [26] (**Reported**).

Top recommendations

1. Run the August 2026 builds or later (Subscription Edition 16.0.19725.20522, 2019 16.0.10417.20198, 2016 16.0.5565.1001) and verify by build number, not by CVE list [11] [16] (**Confirmed** builds; build-first verification is our reading of [9] [10]).
2. Remove machine-key harvesters, then rotate IIS machine keys [1]; assess for compromise [13] [14] (**Confirmed**).
3. Take SharePoint off the direct internet or front it with an authenticating Layer 7 reverse proxy; block external Central Administration [1] (**Confirmed**).
4. Enable AMSI per web application, with Full Mode request body scanning where feasible, and alert on the AMSI and Defender detections CISA names [1] (**Confirmed**).
5. Move off SharePoint Server 2016 and 2019 [15] [17] (**Confirmed**).

Scope and method

This brief covers the eight 2026 SharePoint Server CVEs in KEV, from Microsoft's January fix for CVE-2026-20963 to the KEV catalog of 10 November 2026. The 2025 wave is covered in The SharePoint ToolShell Mass Exploitation (2025); SharePoint in Microsoft 365 is out of scope. The Year 2026 in Intrusions summarizes the wave in one paragraph; this brief is the home study for it. Tradecraft is described at advisory level only.

Grade	Meaning in this study
Confirmed	Stated in an opened primary: MSRC, KEV, CISA, CERT-EU, Canadian Cyber Centre, Microsoft Lifecycle
Reported	Journalism or vendor research, opened but not independently verified
Demonstrated	Seen in a researcher's own honeypot (Previdian), or in the library's own data and studies

Grade	Meaning in this study
Inferred	This study's own reasoning from cited facts

watchTowr, Defused, Viettel and Shadowserver are known only as relayed.

Part 1: what happened

- **13 Jan to 18 Mar.** Microsoft fixes CVE-2026-20963 as not exploited [5]; on 17 March rev 1.1 raises it to an unauthenticated 9.8 [8], and KEV adds it [3] (**Confirmed**); the press noted Microsoft had not marked it exploited [19] [20] (**Reported**). CERT-EU 2026-004 follows on 25 March [13] (**Confirmed**).
- **14 Apr.** Microsoft fixes CVE-2026-32201 as Exploitation Detected; KEV adds it the same day [3] [6] (**Confirmed**).
- **21 and 26 May.** Microsoft publishes CVE-2026-45659, then says its fix shipped in the May updates [9] (**Confirmed**).
- **June.** The CVE-2026-58644 fix ships in the June updates without its CVE [10] (**Confirmed**).
- **14 and 15 Jul.** July updates list CVE-2026-56164 (exploited), 58644, 50522 and 55040; CISA issues its hardening alert [1] [2] [7]. Next day MSRC rev 1.2 marks CVE-2026-58644 exploited, Canada issues AL26-017, and 2016 and 2019 reach end of extended support [10] [15] [17] (**Confirmed**).
- **16 Jul to 26 Aug.** CISA revises its alert on 16 Jul, 28 Jul, 18 Aug and 26 Aug to record KEV additions and, last, to clarify the guidance on avoiding direct internet exposure [1] [3] (**Confirmed**).
- **20 and 22 Jul.** CERT-EU reports that watchTowr found public proof-of-concept code for CVE-2026-50522 and then observed active exploitation; Defused says its honeypot hits were likely that CVE [14] [21] (**Confirmed** for CERT-EU's report; **Reported** for Defused). CERT-EU issues 2026-009 on 22 July [14] (**Confirmed**).
- **22 to 31 Jul.** A Warlock intrusion runs from SharePoint web shell to ransomware [25] [26] (**Reported**).
- **11 and 27 Aug.** August updates fix CVE-2026-65660 [11] (**Confirmed**), and the KEV ransomware flag for CVE-2026-45659 is reported as changed to Known [23] (**Reported**). MSRC rev 1.1 re-titles CVE-2026-65660 as remote code execution [11] (**Confirmed**).
- **24 and 25 Sep.** Previdian's honeypot records a two-stage exploitation attempt [18] (**Demonstrated**); Canada issues AL26-023 [16] (**Confirmed**). Microsoft states (rev 1.2) that it had reliable evidence of observed attacks on CVE-2026-65660 as of 25 September; KEV adds it [3] [4] [11] (**Confirmed**).
- **1 Oct.** Symantec reports Warlock attacks on at least four organizations after SharePoint exploitation [25] (**Reported**).

Discrepancies in the record

- **What CISA called exploited.** The 14 July bulletin named three CVEs as exploited [2]; the revised alert states exploitation of six [1] (**Confirmed**).
- **Privileges.** CVE-2026-50522 carries a no-privileges vector while its FAQ says Site Owner [7], and CERT-EU notes findings that exploitation may not need authentication [14] (**Confirmed**).

- **End-of-support date.** Microsoft Lifecycle gives 15 July 06:59:59 PT; AL26-017 says 14 July [15] [17] (**Confirmed**).
- **Scores and fix month.** CERT-EU 2026-009 lists CVE-2026-56164 at CVSS 9.8 and CVE-2026-58644 as fixed in July; Microsoft gives 5.3 and a June fix [7] [10] [14] (**Confirmed**).

Part 2: the technical record

CVE	Class (CWE)	CVSS, privileges	Fixed	Microsoft exploited flag, 11 Nov	KEV added, due
CVE-2026-20963	RCE (CWE-502)	9.8 after 17 Mar, unauthenticated	13 Jan	No	18 Mar, 21 Mar
CVE-2026-32201	Spoofing (CWE-20)	6.5, unauthenticated	14 Apr	Yes	14 Apr, 28 Apr
CVE-2026-45659	RCE (CWE-502)	8.8, Site Member	May updates (CVE 21 May)	No	1 Jul, 4 Jul
CVE-2026-58644	RCE (CWE-502)	9.8, unauthenticated	June updates (CVE 14 Jul)	Yes (from 15 Jul)	16 Jul, 19 Jul
CVE-2026-56164	EoP (CWE-306)	5.3 Moderate, unauthenticated	14 Jul	Yes	14 Jul, 17 Jul
CVE-2026-50522	RCE (CWE-502)	9.8, vector PR:N, FAQ Site Owner	14 Jul	No	22 Jul, 25 Jul
CVE-2026-55040	Feature bypass (CWE-1390)	9.1, unauthenticated	14 Jul	No	18 Aug, 21 Aug
CVE-2026-65660	Code injection (CWE-94)	8.8, authenticated low privilege	11 Aug	Revision cites attacks from 25 Sep	25 Sep, 28 Sep

Sources: [3] [5] to [11] [15] [16] (**Confirmed**). Credits: CVE-2026-50522 to DEVCORE via the Zero Day Initiative [7]; CVE-2026-45659 to MEOW [9] and CVE-2026-65660 to khoadha [11]; CVE-2026-56164 to Mandiant Incident Response and Google Cloud FLARE OTF researchers, released already exploited [12] (**Confirmed**), so probably found during an intrusion investigation (**Inferred**).

The vendor. Microsoft's record moved after release: CVE-2026-45659 was published on 21 May although its fix was in the May updates, and CVE-2026-58644 on 14 July although its fix was in the June updates [9] [10]; CVE-2026-58644's exploited flag was corrected the day after release [10]; and CVE-2026-20963 and CVE-2026-65660 were re-scored and re-titled [8] [11] (**Confirmed**). The August listing still carries 2016 and 2019 fixes [11] (**Confirmed**).

After getting in, at advisory level. CISA describes theft of IIS machine keys and the use of deserialization to gain persistence and deploy malware [1] (**Confirmed**). Stolen machine keys were reported to let attackers forge valid authentication tokens and keep access after patching [22] (**Reported**).

The September attempt. On 24 September Previdian's honeypot saw a first stage that weakened a .NET deserialization safeguard, then an encrypted assembly loader; Previdian pairs CVE-2026-65660 with an anonymous-delivery bug and calls it an attempted pre-authentication chain, not a compromise [18] (**Demonstrated**). Canada's AL26-023 says such chaining enables pre-authentication code execution on

servers that allow anonymous access [16] (**Confirmed**).

Detection content. The AMSI and Defender detections to alert on are those in CISA's alert, as in Top recommendation 4 [1] (**Confirmed**).

Part 3: who and why

No government record opened names an actor [1] [13] [15] [16], and the KEV ransomware flag names no gang [3] (**Confirmed**).

Symantec's Longlegs and Warlock. Symantec reports that Warlock ransomware, run by a group it calls Longlegs and describes as China-nexus, hit at least four organizations in the two months before 1 October: a water utility, a telecom provider, a regional government body and a university [25] (**Reported**). Initial access was on-premises SharePoint exploitation, then a web shell and ASP.NET machine-key harvesting; Symantec says the 2025 ToolShell CVEs likely remain in the toolkit and ties no 2026 CVE to any intrusion [25] (**Reported**).

Microsoft's Storm-2603. Microsoft DART says Storm-2603 has targeted on-premises SharePoint since mid-2025, naming no CVE [27] (**Reported**). Kaspersky attributes a fileless SharePoint breach to Storm-2603 with moderate confidence, using ToolShell CVE-2025-53770 and CVE-2025-53771 on 24 December (late 2025, by the report's context) [28] (**Reported**). That is ToolShell lineage, not the 2026 wave (**Inferred**).

Is Longlegs Storm-2603? A CyberScoop op-ed treats them as one group; BleepingComputer explicitly does not [26] [31] (**Reported**). This study names them separately.

Lineage. ATT&CK campaign C0058 SharePoint ToolShell Exploitation records the July 2025 wave and names Storm-2603, Threat Group-3390 (G0027) and ZIRCONIUM (G0128) [32] (**Demonstrated**). No record opened ties G0027 or G0128 to a 2026 CVE (**Inferred**). The wider record for these groups and for Storm-2603 is in Threat Actors: China.

Part 4: impact and response

Victims. No primary names a victim. In the sources opened for this study, Symantec's four organizations are the only described victims tied to SharePoint exploitation in 2026, and their CVE is unknown [25] (**Reported**). In the 22 to 31 July intrusion, the attackers disabled security tools on at least 40 hosts in about two hours, staged Warlock in SYSVOL and launched it on at least 33 hosts, using a VS Code tunnel installed as a service and a vulnerable signed driver [25] [26] (**Reported**).

Exposure. Relayed Shadowserver counts were more than 10,000 internet-facing servers in July [29] and more than 8,500 in August, over 200 unpatched against CVE-2026-45659 [23] (**Reported**). These are reachable servers, not compromises (**Inferred**).

CISA and KEV. Every KEV deadline was three days except CVE-2026-32201's 14 days; the CVE-2026-45659 entry cites BOD 26-04 [3] (**Confirmed**).

National CERTs. CERT-EU's 2026-004 (March) recommends updating, AMSI Full Mode, EDR, machine-key rotation and a compromise assessment [13]; 2026-009 (July) adds credential rotation for exposed assets and says organisations should reconsider exposing SharePoint directly to the internet [14]. Canada's AL26-017 advised moving off 2016 and 2019 [15], and AL26-023 reported active exploitation of CVE-2026-65660 [16]

(**Confirmed**).

Part 5: the defender and vulnerability-management view

Fixes arrived in builds before their CVEs were listed, and exploitation arrived in KEV while Microsoft's flag stayed at No [3] [9] [10] (**Confirmed**). A program that patched only what Microsoft marks exploited would still be waiting on four CVEs as of 11 November 2026 (**Inferred**).

Control	Where it meets the record
Patch by build, then verify the build	Silent May and June fixes [9] [10]; CISA [1]
Remove harvesters, rotate keys, assess	CISA [1]; CERT-EU [13] [14]
No direct internet exposure	CISA [1]
AMSI per web application, Full Mode where feasible	CISA [1]; MSRC mitigating factor for CVE-2026-56164 [7]
Disable anonymous access where it is not needed	Chaining needs anonymous access [16] [18] (**Inferred** control)
Tamper protection and driver blocking	Warlock operators disabled tools with a vulnerable driver [26] (**Reported**)

What is overstated and what is understated

- **Overstated:** any link from Warlock or Storm-2603 to CVE-2026-45659 or its successors goes beyond Symantec's report, which names no 2026 CVE [25] (**Reported**). Equating Longlegs with Storm-2603 is disputed [26] (**Reported**). Exposure counts measure reachable servers, not compromises [23] [29] (**Inferred**). Resecurity's "Operation FarmKey" is a fictional example [30] (**Reported**).
- **Understated:** Microsoft's exploited flags for four KEV-listed CVEs [3] [7] [8] [9] (**Confirmed**); the two silent fixes [9] [10] (**Confirmed**); CVE-2026-56164's Moderate 5.3 score next to Exploitation Detected, against CERT-EU's 9.8 [7] [14] (**Confirmed**). Four victims is a floor, not a total (**Inferred**).

MITRE mappings

Names and tactics come from the library's ATT&CK v20.0, CWE, CAPEC and D3FEND data [32] (**Demonstrated**). No source names ATT&CK ids, so every id is this study's mapping (**Inferred**). No source describes industrial-control activity, so no ATT&CK for ICS techniques apply [25].

Groups, software and campaigns

ATT&CK has no object for Storm-2603, Longlegs or Warlock [32] (**Demonstrated**). Lineage only: campaign C0058 SharePoint ToolShell Exploitation (see the ToolShell study [33]) and the groups it names, G0027 Threat Group-3390 and G0128 ZIRCONIUM [32].

ATT&CK Enterprise techniques

ID	Name	Tactic(s)	Dated event or report	Sources
T1190	Exploit Public-Facing Application	Initial Access	Eight CVEs in KEV, 18 Mar to 25 Sep; CISA alert, 14 Jul	[1] [3]
T1552.001	Credentials In Files	Credential Access	Machine-key theft (CISA, 14 Jul; Symantec, 1 Oct)	[1] [25]
T1606	Forge Web Credentials	Credential Access	Token forgery with stolen keys, reported 21 Jul	[22]
T1105	Ingress Tool Transfer	Command and Control	Malware deployed via deserialization (CISA, 14 Jul)	[1]
T1505.003	Web Shell	Persistence	Web shells (Symantec, 1 Oct); attempts, 25 Sep	[24] [25]
T1620	Reflective Code Loading	Stealth	Encrypted assembly loader attempt (Previdian, 24 Sep)	[18]
T1685	Disable or Modify Tools	Defense Impairment	Tools disabled on 40 hosts, 22 to 31 Jul	[26]
T1068	Exploitation for Privilege Escalation	Privilege Escalation	Vulnerable signed driver, 22 to 31 Jul	[26]
T1543.003	Windows Service	Persistence, Privilege Escalation	Tunnel installed as a service, 22 to 31 Jul	[26]
T1219.001	IDE Tunneling	Command and Control	VS Code tunnel, 22 to 31 Jul	[26]
T1570	Lateral Tool Transfer	Lateral Movement	Warlock staged in SYSVOL, 22 to 31 Jul	[26]
T1486	Data Encrypted for Impact	Impact	Warlock on 33 hosts, 22 to 31 Jul	[25] [26]

Fit notes (**Inferred**): C0058 also lists T1552.001 [32]; T1606 is a reported capability and T1620 an inexact fit; the Warlock rows' CVE is unknown [25].

CWE and CAPEC

CWE ids as MSRC and KEV list them: CWE-502 (CVE-2026-20963, -45659, -58644, -50522), CWE-20 (-32201), CWE-306 (-56164), CWE-1390 (-55040) and CWE-94 (-65660) [3] [5] to [11] (**Confirmed**).

CAPEC	Name	Via	Where it meets the record
CAPEC-586	Object Injection	CWE-502	The four deserialization CVEs [3]
CAPEC-242	Code Injection	CWE-94	CVE-2026-65660 [11]
CAPEC-36	Using Unpublished Interfaces or Functionality	CWE-306	CVE-2026-56164 [7] (**Inferred** fit)

The library relates no CAPEC to CWE-1390 and more than fifty to CWE-20, so neither is bound [32] (**Demonstrated**). See the CWE Reference and CAPEC Reference.

D3FEND countermeasures

D3FEND countermeasure	Where it meets the record
Inbound Traffic Filtering	Layer 7 reverse proxy [1]
Network Traffic Filtering	Restrict Central Administration, farm and database traffic [1]
Application Protocol Command Analysis	AMSI request body scanning [1] [7] (**Inferred** fit)
File Integrity Monitoring	Hunting web shells and harvesters [1] (**Inferred** fit)
Credential Rotation	Rotate IIS machine keys [1] [13] [14]
Credential Compromise Scope Analysis	CERT-EU's compromise assessment [13] [14] (**Inferred** fit)

See the D3FEND Reference.

Navigator layer

The techniques are saved as an ATT&CK Navigator layer, `navigator/library-layers/sharepoint-2026-exploitation.json` (open in Navigator), listed on the Navigator layers page, pinned to ATT&CK v20, `enterprise-attack`. Scores: 75 government or KEV, 50 vendor or researcher report, 25 capability, attempt or inexact fit; each comment names its sources.

Cross-cutting themes

- **End of support during an active wave.** 2016 and 2019 left support the day after the July release that listed four SharePoint CVEs, one of them already fixed in June [7] [10] [17] (**Confirmed**).
- **Attribution stays thin.** Eight KEV entries and no government-named actor [1] [3] [15] [16] (**Confirmed**).

Sources

All sources were opened on 11 November 2026.

#	Source	URL	Opened	Grade
1	CISA, CISA Urges SharePoint Hardening After New Exploitations, 14 Jul 2026, last revised 26 Aug 2026	cisa.gov/news-events/alerts/2026/07/14/...	2026-11-11	Confirmed
2	CISA GovDelivery bulletin reproducing the 14 Jul 2026 alert	content.govdelivery.com/.../bulletins/4208511	2026-11-11	Confirmed
3	CISA, Known Exploited Vulnerabilities catalog JSON, version 2026.11.10	cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-11	Confirmed
4	CISA GovDelivery, CISA Adds Two Known Exploited Vulnerabilities to Catalog, 25 Sep 2026	content.govdelivery.com/.../bulletins/42c823a	2026-11-11	Confirmed
5	MSRC CVRF, January 2026 security updates	api.msrc.microsoft.com/cvrf/v3.0/cvrf/2026-Jan	2026-11-11	Confirmed

#	Source	URL	Opened	Grade
6	MSRC CVRF, April 2026 security updates	api.msrc.microsoft.com/cvrf/v3.0/cvrf/2026-Apr	2026-11-11	Confirmed
7	MSRC CVRF, July 2026 security updates	api.msrc.microsoft.com/cvrf/v3.0/cvrf/2026-Jul	2026-11-11	Confirmed
8	MSRC Security Update Guide, CVE-2026-20963	api.msrc.microsoft.com/sug/v2.0/en-US/vulnerability/CVE-2026-20963	2026-11-11	Confirmed
9	MSRC Security Update Guide, CVE-2026-45659	api.msrc.microsoft.com/sug/v2.0/en-US/vulnerability/CVE-2026-45659	2026-11-11	Confirmed
10	MSRC Security Update Guide, CVE-2026-58644	api.msrc.microsoft.com/sug/v2.0/en-US/vulnerability/CVE-2026-58644	2026-11-11	Confirmed
11	MSRC Security Update Guide, CVE-2026-65660	api.msrc.microsoft.com/sug/v2.0/en-US/vulnerability/CVE-2026-65660	2026-11-11	Confirmed
12	MSRC Security Update Guide, acknowledgements	api.msrc.microsoft.com/sug/v2.0/en-US/acknowledgement	2026-11-11	Confirmed
13	CERT-EU, Security Advisory 2026-004, 25 Mar 2026	cert.europa.eu/publications/security-advisories/2026-004/	2026-11-11	Confirmed
14	CERT-EU, Security Advisory 2026-009, v1.0 and v1.1, 22 Jul 2026	cert.europa.eu/publications/security-advisories/2026-009/	2026-11-11	Confirmed
15	Canadian Centre for Cyber Security, AL26-017, 15 Jul 2026	cyber.gc.ca/en/alerts-advisories/al26-017-...	2026-11-11	Confirmed
16	Canadian Centre for Cyber Security, AL26-023, 24 Sep 2026	cyber.gc.ca/en/alerts-advisories/al26-023-...	2026-11-11	Confirmed
17	Microsoft Lifecycle, SharePoint Server 2019	learn.microsoft.com/en-us/lifecycle/products/sharepoint-server-2019	2026-11-11	Confirmed
18	Previdian, CVE-2026-65660: Previdian observes two-stage SharePoint exploitation attempts, 24 Sep 2026, updated 25 Sep 2026	blog.previdian.com/cve-2026-65660-...	2026-11-11	Demonstrated
19	The Register, report on unknown attackers exploiting CVE-2026-20963, 19 Mar 2026	theregister.com/2026/03/19/unknown_attackers_exploit_yet_another/	2026-11-11	Reported
20	Help Net Security, report on CVE-2026-20963 exploitation, 19 Mar 2026	helpnetsecurity.com/2026/03/19/...	2026-11-11	Reported
21	SecurityWeek, report on a fourth SharePoint vulnerability exploited, 22 Jul 2026	securityweek.com/fourth-sharepoint-vulnerability-...	2026-11-11	Reported
22	BleepingComputer, report on SharePoint RCE exploited to steal machine keys, 21 Jul 2026	bleepingcomputer.com/news/security/critical-sharepoint-rce-...	2026-11-11	Reported

#	Source	URL	Opened	Grade
23	BleepingComputer, report on CVE-2026-45659 in ransomware attacks, 11 Aug 2026	bleepingcomputer.com/news/security/cisa-microsoft-sharepoint-...	2026-11-11	Reported
24	SecurityWeek, report on CVE-2026-65660 exploitation, 27 Sep 2026	securityweek.com/microsoft-sharepoint-flaw-cve-2026-65660-...	2026-11-11	Reported
25	Symantec Threat Hunter Team, Warlock ransomware and critical infrastructure, 1 Oct 2026	security.com/threat-intelligence/warlock-ransomware-critical-infrastructure	2026-11-11	Reported
26	BleepingComputer, report on Warlock breaches via SharePoint, 2 Oct 2026	bleepingcomputer.com/news/security/warlock-ransomware-...	2026-11-11	Reported
27	Microsoft Security blog (DART), One intrusion, two cyberattackers, 22 Jun 2026	microsoft.com/en-us/security/blog/2026/06/22/...	2026-11-11	Reported
28	Kaspersky Securelist, Storm-2603 fileless ToolShell attack, 5 Nov 2026	securelist.com/storm-2603-fileless-toolshell-attack/121467/	2026-11-11	Reported
29	Cloud Security Alliance, research note on CVE-2026-45659, 3 Jul 2026	labs.cloudsecurityalliance.org/research/...	2026-11-11	Reported
30	Resecurity, understanding the July 2026 SharePoint attacks, 18 Jul 2026	resecurity.com/blog/article/...	2026-11-11	Reported
31	CyberScoop, op-ed on Warlock and SharePoint attacks, 27 Oct 2026	cyberscoop.com/warlock-ransomware-storm-2603-sharepoint-attacks-op-ed/	2026-11-11	Reported
32	TeamStarWolf library data: `data/attack/technique_profiles.jsonl`, `campaigns.jsonl`, `campaign_profiles.jsonl`, `groups.jsonl`, `software.jsonl`, `technique_to_d3fend.jsonl` (ATT&CK v20.0), `data/weaknesses/cwe.jsonl` and `capec.jsonl`, and the `mitre/` page tree including C0058	github.com/TeamStarWolf/TeamStarWolf/tree/main/data/attack	2026-11-11	Demonstrated
33	TeamStarWolf, The SharePoint ToolShell Mass Exploitation: A 2025 Case Study	/case-studies/SHAREPOINT_TOOLSHELL_2025.md	2026-11-11	Demonstrated

Not opened, so not cited as sources: watchTowr's and Defused's own posts, Viettel Cyber Security's technical write-up, the public proof-of-concept repository and Shadowserver's dashboards.

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Related references

- The SharePoint ToolShell Mass Exploitation (2025), Threat Actors: China, The Ransomware Ecosystem, The Year 2026 in Intrusions, Timeline
- Threat Actors, Malware Families, Notable Incidents
- CVE Reference, CWE Reference, CAPEC Reference, D3FEND Reference
- Web Application Security Reference, Ransomware Defense Reference, Windows Hardening Reference
- Vulnerability Management Reference, Vulnerability Prioritization Reference
- Campaigns: C0058 SharePoint ToolShell Exploitation
- Groups: G0027 Threat Group-3390, G0128 ZIRCONIUM
- Techniques: T1190 Exploit Public-Facing Application, T1552.001 Credentials In Files, T1606 Forge Web Credentials, T1105 Ingress Tool Transfer, T1505.003 Web Shell, T1620 Reflective Code Loading, T1685 Disable or Modify Tools, T1068 Exploitation for Privilege Escalation, T1543.003 Windows Service, T1219.001 IDE Tunneling, T1570 Lateral Tool Transfer, T1486 Data Encrypted for Impact
- CAPEC: CAPEC-586, CAPEC-242, CAPEC-36
- Navigator layers, including this study's layer, sharepoint-2026-exploitation.json
- The Team Star Wolf Intelligence Hub
- All case studies